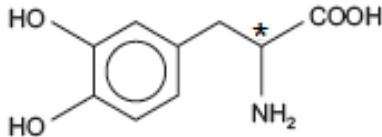
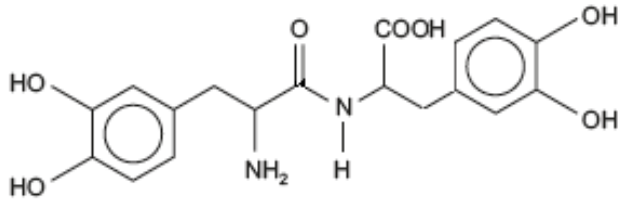
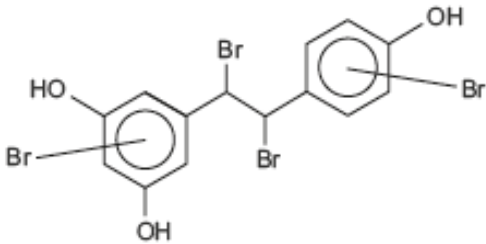


Question				Marking details	Marks available								
					AO1	AO2	AO3	Total	Maths	Prac			
12.	(a)			Indicative content 65.2 % of carbon and M_r 184 'M_r' of the carbon content is $65.2 \times 184 / 100 = 120$ therefore must be 10 carbon atoms 26.1 % of oxygen and M_r 184 'M_r' of the oxygen content is $26.1 \times 184 / 100 = 48$ therefore must be 3 oxygen atoms remaining mass 16 therefore must be 16 hydrogen atoms $\Rightarrow$ molecular formula is $C_{10}H_{16}O_3$ straight chain, double bond between carbons 2 and 3 and an <i>E</i>-isomer $\begin{array}{ccccccccccc} & & & & & & & & & & \\ & & & & & & & & C & & \\ C & - & C & - & C & - & C & - & C & - & C & - & C & = & C & - & C \\ & & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & & \end{array}$ 2,4-DNP derivative, but no silver mirror therefore must be a ketone gives the triiodomethane test therefore must contain $CH_3C=O$ group; these must be carbon atoms 9 and 10 effervescence therefore must be a carboxylic acid; not branched, therefore carbon 1 must be the carboxylic acid group formula must be $\begin{array}{ccccccccccccccc} & H & & O & & H & & H & & H & & H & & H & & H & & & O \\ & & & & & & & & & & & & & & & & & & \\ H & - & C & - & C & - & C & - & C & - & C & - & C & - & C & = & C & - & C \\ & & & & & & & & & & & & & & & & & & \backslash \\ & H & & & & H & & H & & H & & H & & H & & H & & & OH \end{array}$									

			5-6 marks Correct structure given; reference to all information provided <i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks Molecular formula obtained but insufficient chemical deductions to obtain the correct structure <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks Some use of analytical results and / or chemical deductions but insufficient to suggest a structure <i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						
	(b)	(i)			1		1		
		(ii)	one isomer rotates the plane to the left and the other to the right (1) isomers present in equimolar amounts therefore the effect of rotation is cancelled out (1)	2			2		2
		(iii)			1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		benzene compounds are very resistant to addition reactions and this would destroy the stable ring system of electrons / needs too much energy	1			1		
		(ii)		 at least one Br needed on both rings (any position)			1	1		
				Question 12 total	3	4	5	12	0	5